RAMCO INSTITUTE OF TECHNOLOGY
Approved by AICTE and Affiliated to Anna University,
Accredited by NAAC, AN ISO 9001:2015 Certified)

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

# ABOUT THE DEPARTMENT

The Department of Electronics and Communication Engineering was established in the year 2013 with a vision to develop competent professionals to meet the global challenges in modern engineering society. At present it offers a 4-year B.E. Electronics and Communication Engineering programme with a sanctioned intake of 120 students. The Laboratories are equipped with latest technology in the fields of Electronic Devices and Circuits, Digital Electronics, Linear Integrated Circuits, Microprocessors and Microcontrollers, Digital Signal Processing, Communication System, VLSI, Embedded Systems, Optical & Microwave, Embedded systems that offer opportunities on a wide range of hardware and advanced software packages. In addition to this, the Department has equipped with Research Laboratories such as NI-LabVIEW Research Laboratory, Tessolve Semiconductor Laboratory and Technical Incubation Centre to enrich the knowledge of students and faculty. The Department has a team of committed faculty members with sound knowledge in the areas of Applied Electronics, Communication systems, VLSI, Embedded Systems, Computer Communication and Networking. The Department encompasses professional associations such as Students Technical Association Electronica, IETE, ISTE chapters which provide students the platform to learn professional and soft skills. The Department delights to release the Technical Magazine - "ELECOM MAGAZINSIO" for the academic year 2022-2023

## VISION

To evolve as an Institute of international repute in offering high- quality technical education, Research and extension programmes in order to create knowledgeable, professionally competent and skilled Engineers and Technologists capable of working in multi- disciplinary environment to cater to the societal needs.

## MISSION

- To accomplish its unique vision, the Institute has a far-reaching mission that aims:
- To offer higher education in Engineering and Technology with highest level of quality, professionalism and ethical standards
- To equip the students with up- to- date knowledge in cutting- edge technologies, wisdom, creativity and passion for innovation, and life-long learning skills
- To constantly motivate and involve the students and faculty members in the education process for continuously improving their performance to achieve excellence

## QUALITY POLICY

RIT aims to impart Quality Education in Engineering and Technology through an effective teaching- learning process, up- gradation of facilities and human resources, collaborating with Industry for promoting training and placement, research and consultancy activities with a commitment to continual improvement of Quality Management System.

## VISION

Develop futuristic technologist in Electronics and Communication Engineering to meet the global challenges and the societal needs of our nation.

## MISSION

- Educate students to become enlightened engineers with the latest know-how to take on the challenges concerned with the society.
- Equip the students with the current trends and latest technologies in the field of Electronics & Communication Engineering to make them technically strong and ethically sound.
- Constantly motivate the students and faculty members to achieve excellence in Electronics & Communication Engineering and Research & Development activities.

## PROGRAM EDUCATIONAL OBJECTIVES (PEOs) :

- Graduates will gain knowledge through continuous learning to enhance competency and career prospects.
- Graduates will be technically competent to design, analyze, develop and implement Electronic and Communication Systems.
- Graduates will apply the basic fundamental concepts in Electronic and related fields to solve general engineering problems related to societal needs.
- Graduates will be professionals with attributes, passion, positive attitude, ethics and moral values for a successful career.
- Graduates will Communicate effectively and interact responsibly with colleagues, employees, team members, clients and the society.

## PROGRAM SPECIFIC OUTCOMES (PSOS) :

After successful completion of the degree, the students will be able to:

- Adapt to emerging trends in Information and Communication Technology by innovating new ideas to solve existing/novel problems.
- Excel in latest technologies of Embedded Systems.
- Design, analyze and develop electronic products in the area of VLSI Design and Communication Systems.
- Contribute as Entrepreneurs in building electronic products.

# DIGITAL TWINS

**A digital twin is a digital <u>model</u> of an intended or actual real-world physical product, system, or process that serves as the effectively indistinguishable digital counterpart of it for practical purposes, such as <u>simulation</u>, <u>integration</u>, <u>testing</u>, <u>monitoring</u>, and <u>maintenance</u>. the digital twin has been intended from its initial introduction to be the underlying premise for product lifecycle management and exists throughout the entire lifecycle of the physical entity it represents. since information is granular, the digital twin representation is determined by the value-based use cases it is created to implement. the digital twin can and does often exist before there is a physical entity. the use of a digital twin in the creation phase allows the intended entity's entire lifecycle to be modeled and simulated. a digital twin of an existing entity may be used in real-time and regularly synchronized with the corresponding physical system.**

## HOW IT WORKS?

**A digital twin is a virtual model that aims to accurately reflect a physical object. these sensors generate data about various aspects of physical object performance, such as: b. energy output, temperature, weather conditions, etc. this data is then passed to a processing system and applied to a digital copy. once the virtual model is provided with such data, it can be used to run simulations to investigate performance issues and generate possible improvements. the goal of all of this is to gain valuable insights that can then be applied to the original physical object.**


# ADVANTAGES AND BENEFITS :

**Digital twins improve r&d by providing valuable data for product design and refinement. they also enhance efficiency by monitoring production systems post-production, ensuring peak manufacturing performance. additionally, they aid in determining the end-of-life of products, enabling manufacturers to decide on final processing methods like recycling or other measures, ensuring the sustainability of their products. overall, digital twins enhance the efficiency and effectiveness of manufacturing processes.**

**M.SIVA RANJANI**
**IV – ECE B**


5G technology represents the fifth generation of mobile network technology, promising a significant leap forward in terms of speed, connectivity, and overall performance compared to its predecessors. The key features of 5G include faster data transfer rates, lower latency, and the ability to connect a massive number of devices simultaneously. With data speeds potentially reaching up to 100 times faster than 4G, 5G opens the door to transformative possibilities in various industries, such as healthcare, transportation, and manufacturing. The reduced latency in 5G networks allows for near-instantaneous communication, enabling advancements in technologies like autonomous vehicles and remote surgery. Additionally, 5G facilitates the Internet of Things (IoT) by providing a robust and reliable connection for an extensive array of devices. While 5G brings exciting opportunities, its deployment also raises concerns about privacy, security, and the potential environmental impact of increased energy consumption. Despite these challenges, 5G technology is poised to revolutionize the way we live, work, and communicate in the years to come.


## Uses

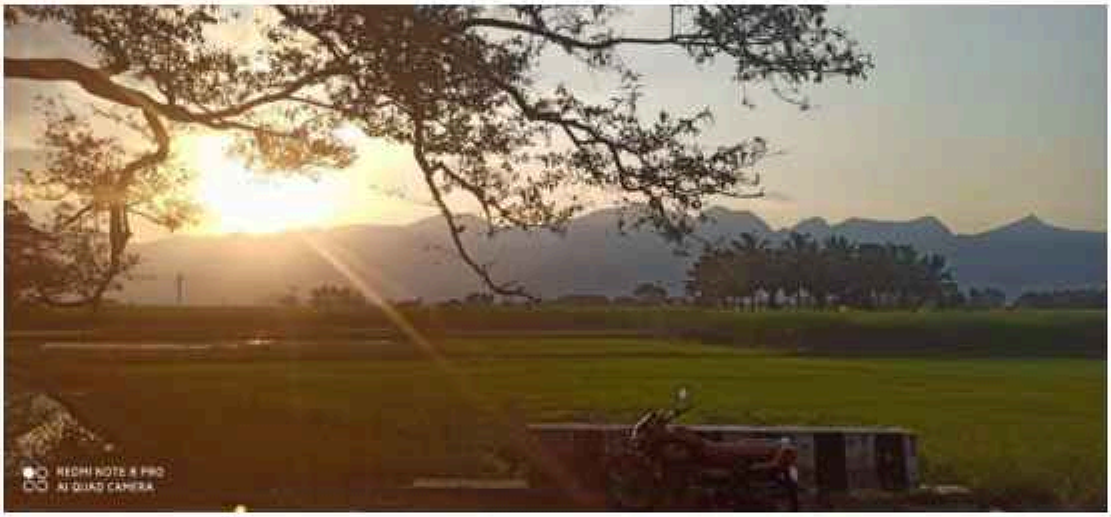
- **Enhanced Mobile Broadband (eMBB):** 5G provides significantly faster data speeds, enabling users to experience high-definition video streaming, virtual reality (VR), and augmented reality (AR) applications with minimal latency. This enhances the overall mobile broadband experience for consumers.
- **Internet of Things (IoT):** 5G's ability to handle a massive number of connected devices simultaneously makes it a crucial enabler for the IoT. It allows for seamless communication and data exchange between devices, fostering the growth of smart homes, smart cities, and industrial IoT applications.
- **Education:** 5G enhances educational experiences by supporting high-quality, immersive content for remote learning. It enables interactive virtual classrooms, online collaboration, and access to educational resources with minimal latency.

## Advantages :

- **High Speeds:** 5G offers faster data transfer rates.
- **Low Latency:** Minimal delay in data transmission.
- **Massive Connectivity:** Supports numerous devices simultaneously.
- **IoT Enablement:**Crucial for the Internet of Things.
- **Revolutionizes Industries:** Enhances healthcare, autonomous vehicles, and more.

AnusuyaG
IV YEAR ECE - A

# Photography



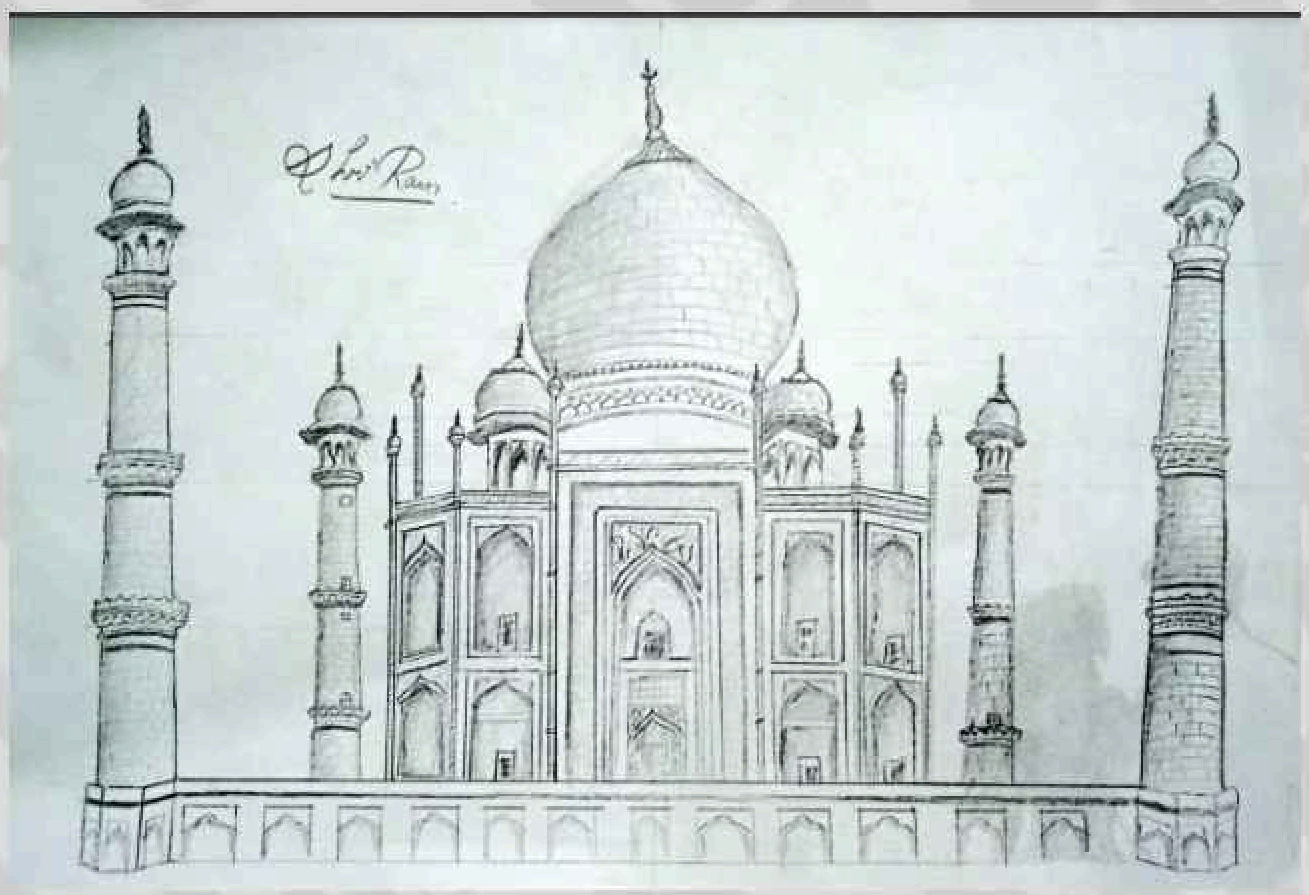

Sudhaneswari P

IV YEAR ECE - B

Sri Nandhini Priya S

IV YEAR ECE - B

# PeNCil sketch



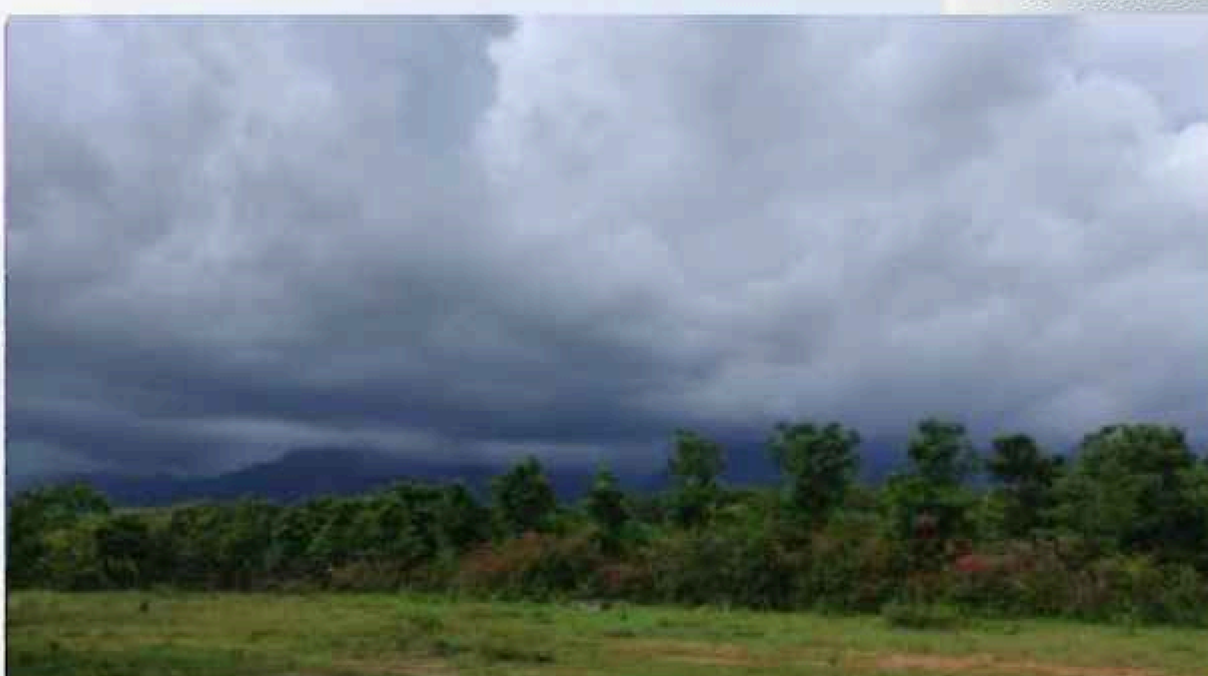


**Shriram**
II YEAR ECE

# Photography

Monica S

IV YEAR ECE B



ShakthiSreeMA

IV YEAR ECE B

# POWER BI

Power BI (Business Intelligence) is a suite of business analytics tools developed by Microsoft. It allows users to connect to various data sources, analyze and visualize data, and share insights across an organization. Power BI consists of components like Power BI Desktop (for creating reports and dashboards), Power BI Service (cloud-based platform for sharing and collaborating on reports), and Power BI Mobile (for accessing reports on mobile devices). The tool is widely used for data modeling, transforming raw data into meaningful insights, and creating interactive and customizable reports and dashboards for informed decision-making in businesses.


## Difference between POWER BI and EXCEL :

Power BI and Excel serve distinct purposes in data analysis. Excel is a spreadsheet tool primarily used for tabular data and calculations, while Power BI is a comprehensive business intelligence platform designed for data visualization, analysis, and sharing insights across organizations, offering more advanced capabilities for interactive reporting and dashboard creation.

Shakthi Sree MA
IV YEAR ECE - B

## Pencil sketch

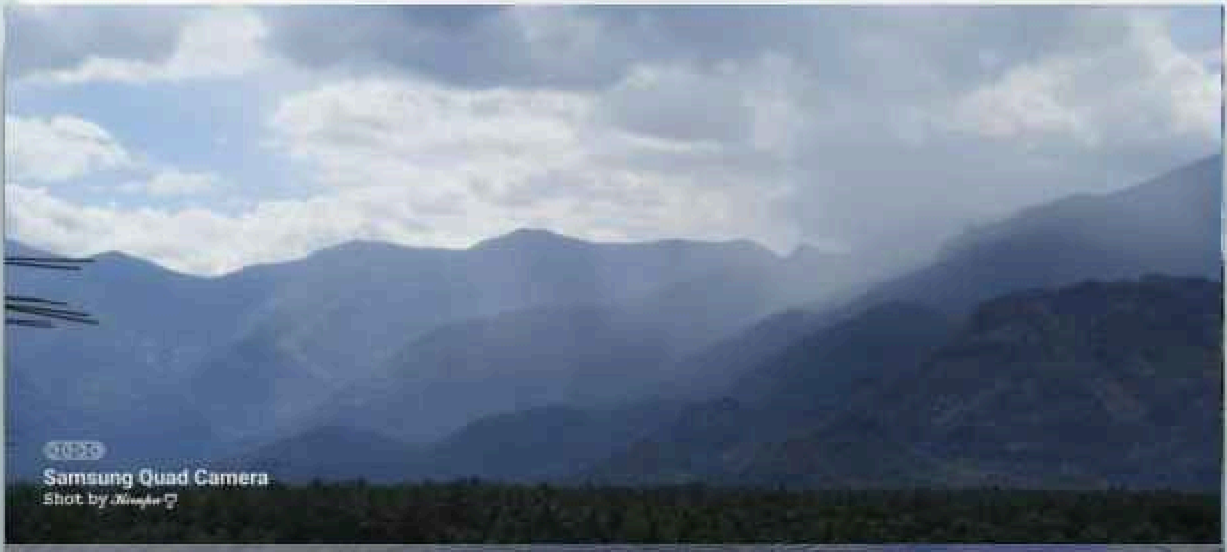
Siva Ranjani M

IV YEAR ECE B

SuruthiS

IV YEAR ECE B

# PHOTOGRAPY




Kriupa M

IV YEAR ECE A

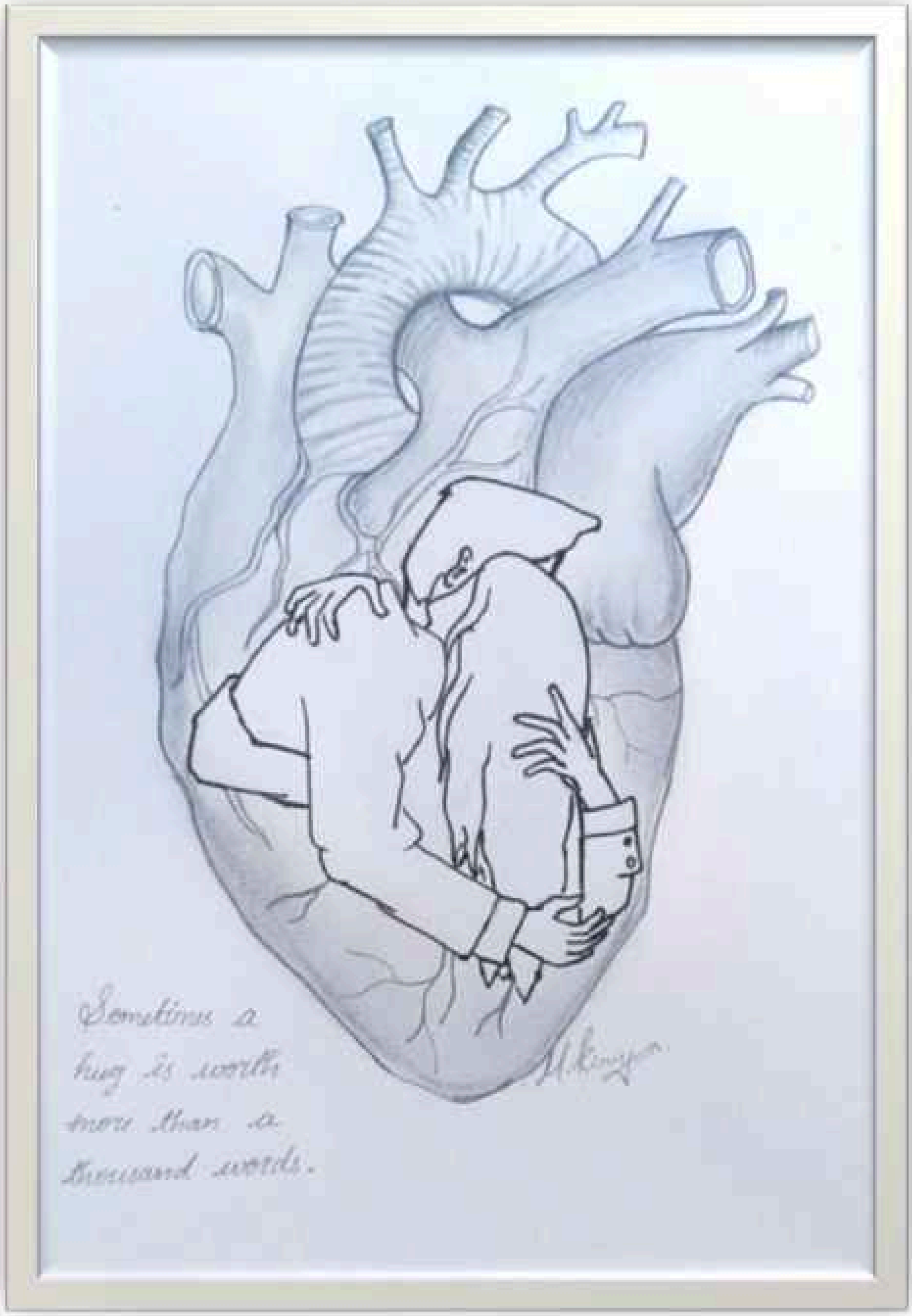
# பள்ளி

பள்ளியின் முதல்நாள், அன்னையின் கையைவிட்டு
ஆசிரியர் கைப்பிடித்த எனக்கு, அன்னைக்கு
நிகரான அன்பை உணர்த்தியது, உனது விரல்கள்
எனது கண்ணீரை துடைத்த நொடி....
ஆசிரியர் பாடம் நடத்தும் வேளை,
வர்ணிக்க முடியாத உறக்கம் என்னைத்தாக்க
அதை உடைத்து எரியும், காகிதத்தை சுருட்டி
என்காதில் நீ நுழைக்கும் அந்த வேளை....
தேர்வு அறைக்குள் நுழையும் பொழுது
நல்ல மதிப்பெண்களில் தேர்ச்சி பெறுவேன்
என்ற எண்ணத்தை என்னுள் விதைக்கும் நீ
என் நெற்றியில் இடும் ஒற்றை திருநீறு....
ஆயிரம் நண்பர்கள் வந்தாலும் உனக்கே
முன்னுரிமை என்று சொல்லாமல் சொல்கிறாய்
இன்றோ நீ வேறு கல்லூரியில் படித்தாலும்
காலை வணக்கம் என்று நீயனுப்பும் குறுஞ்செய்தி வாயிலாக....
இதற்கெல்லாம் நன்றி கூறும் வகையில்
என்நாட்குறிப்பில் மட்டுமல்ல இதயத்திலும்
பாதி இடம் உனக்கே - என் அன்பு தோழியே!

Monica S

IV YEAR ECE B

# drawing




**M.Kirupa**

IV YEAR ECE-A

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